Stepper Motor Shield Project Report

PIC32MK0512MCJ064 + TMC2660 + AS5047D CAN Bus and USB-C Interface

June 29, 2026

Project Status

Area	Status
Firmware	**Complete** — v6.2.0 tested and verified at 48 MHz (SPLL, 115200 baud, auto-tune capped)
Hardware testing	**Complete** — NEMA17 + TB6600 characterized at 24V and 31V (11 test rows at 31V)
Schematic	**Complete** — KiCad schematic for PIC32MK + TMC2660 + AS5047D + CAN + USB-C
PCB layout	**In progress** — routing and layout yet to be finished
10 MHz clock migration	**Pending** — all clock-dependent formulas need recalculation and retesting

What Has Been Done

- Firmware v6.2.0 with SPLL 48 MHz clock, 115200 baud serial, PID position control, S-curve and trapezoidal profiles, auto-tune (Ziegler-Nichols relay method), multi-point queue, telemetry stream, and NVM config storage
- Hardware testing at 24V and 31V — 11 test configurations at 31V characterizing MAXV, ACCEL, and PID gain limits
- KiCad schematic capture with all major components placed and wired: PIC32MK0512MCJ064 (TQFP-64) with decoupling, VCAP, oscillator, ICSP; TMC2660 stepper driver (QFP-44) with SPI control, sense resistors, charge pump; AS5047D magnetic encoder (TSSOP-14) in SPI mode; MCP2562 CAN transceiver (SOIC-8); USB-C for power and programming; JST PH connectors for motor and encoder

What Remains

- PCB layout — component placement, trace routing, thermal management, design rule check
- 10 MHz clock migration — recalculate UART baud (U1BRG), Timer2/OC1 step rate, Timer3 ISR period, core tick timing, auto-tune formulas
- Fabrication — generate Gerber files, BOM, pick-and-place
- Assembly and testing at target clock speed

Firmware

Clock Configuration

All testing performed at **48 MHz** SYSCLK using the SPLL from the internal 8 MHz FRC oscillator:

Parameter	Value
FRC oscillator	8 MHz
PLLIDIV	DIV_2 (4 MHz to PLL)
PLLMULT	MUL_24 (96 MHz VCO)
PLLODIV	DIV_2 (48 MHz SYSCLK)
PB clock	48 MHz
Core timer ticks	24,000 / ms
UART baud	115,200 (U1BRG = 25, 0.16% error)
Timer3 ISR	1 kHz (PR3 = 47,999)
Telemetry interval	200 ms (tlm_ticks = 4,800,000 core ticks)

Target production clock: 10 MHz — all formulas need recalculation and testing is pending.

Features

Feature	Details
Position control loop	1 kHz Timer3 ISR — error → profile feed-forward + PID trim → smoother → motor speed
Step generation	OC1 + Timer2 PWM — hardware-generated step pulses, no CPU bit-banging
Encoder feedback	QEI1 — AS5047D/AS5047U ABI, 16384 counts/rev
Motion profiles	S-curve (jerk-limited) and trapezoidal (instant accel)
PID control	Proportional, integral, derivative with anti-windup
Auto-tune	Relay-based Ziegler-Nichols — outputs Kp, Ki, Kd with caps
Multi-point queue	2–32 waypoints with configurable dwell delay
Telemetry	Configurable status stream over UART
NVM config	Auto-save to last flash page on parameter change
Dashboard	Node.js/WebSocket web UI with sliders, tune button, TLM toggle

Firmware Architecture

Component	Peripheral	Rate	Purpose
Position loop	Timer3 ISR	1 kHz	Profile + PID trim + smoother + velocity
Step generation	OC1 + Timer2 PWM	Variable	Hardware step pulses, 50% duty
Encoder	QEI1	Continuous	AS5047U ABI, 16384 counts/rev
UART	UART1	115200 baud	Command + telemetry
NVM flash	Self-write	On change	Config storage (last flash page)
Status stream	Main loop	Configurable	P:E:V:T:F:M:A telemetry

Control Algorithm

```
error = target_pos - encoder_read()

// Profile feed-forward velocity
stop_dist = max_vel^2 / (2 * accel_limit)
if abs(error) > stop_dist:
    profile_vel = max_vel * sign(error)     // cruise
else:
    profile_vel = max_vel * error / stop_dist // decel ramp

// PID trim
pid_trim = (Kp*error + Ki*integral + Kd*derivative) / 100

// Combined, fed into accel/jerk smoother
raw_vel = profile_vel + pid_trim
```

Resolution

Aspect	Value
Motor steps/rev	200 (NEMA17, 1.8 degrees/step)
Microstep setting	1/32
Steps per revolution	6400
Encoder PPR	4096 (AS5047U)
Encoder counts/rev (4x QEI)	16384
Position resolution	360 degrees / 16384 = 0.022 degrees per count
Achieved accuracy	+-1-3 counts = 0.022-0.066 degrees

Motor Speed Formula

```
motor_set_speed(steps_per_sec):
    pr = (PB_CLOCK / 8) / steps_per_sec
```

```
    PR2 = pr - 1
    OC1R = pr >> 1    // 50% duty cycle
```

The Timer2 clock is PB_CLOCK / 8 = 6 MHz at 48 MHz PB. At 115200 baud, the OC1 produces clean step pulses up to approximately 300 kHz.

Serial Protocol

Baud Rate: 115200

Legacy Commands

Command	Action	Range
`T=<pos>`	Move to absolute position	+/- 2M counts
`STOP`	Emergency stop, hold position	-
`HOME`	Set current position = 0 and target = 0	-
`ZERO`	Set current position = 0 (leave target)	-
`CLEAR`	Clear fault, re-init driver	-
`ON`	Enable motor + spin at last speed	-
`OFF`	Disable motor, exit open-loop	-
`CW`	Set direction clockwise	-
`CCW`	Set direction counter-clockwise	-
`SPEED=<n>`	Step rate for open-loop	10-100000
`MAXV=<n>`	Max velocity	>= 1 steps/s
`ACCEL=<n>`	Max acceleration	>= 100 steps/s^2
`JERK=<n>`	Max jerk (S-curve)	>= 1000 steps/s^3
`PROFILE=S`	S-curve smooth transitions	-
`PROFILE=T`	Trapezoidal instant accel	-
`KP=<n>`	Proportional gain	0-99999
`KI=<n>`	Integral gain	0-99999
`KD=<n>`	Derivative gain	0-99999
`TOL=<n>`	At-target tolerance	0-1000 counts
`FT=<n>`	Fault threshold	1-100000 counts
`GET`	One-shot config dump	-
`Q=<pos>,...`	Load queue (2-32 points)	-
`DWELL=<ms>`	Inter-point dwell delay	0-30000 ms
`QSTOP`	Abort queue mid-execution	-

Phase 1 Commands (Colon-Separated)

Command	Action	Example
`m:<speed>:<pos>`	Move to position at given speed	`m:20000:80000`
`en:1`	Enable closed-loop servo	`en:1`
`en:0`	Disable motor (release)	`en:0`

Command	Action	Example
`z`	Set current encoder position as zero	`z`
`i:<mA>`	Set coil current limit	`i:800`
`us:<n>`	Set microstep (1/2/4/8/16/32)	`us:16`
`pid:<kp>:<ki>:<kd>`	Set PID gains	`pid:50:5:10`
`tlm:1:<ms>`	Enable status stream at N ms period	`tlm:1:50`
`tlm:0`	Disable status stream	`tlm:0`
`st?`	One-shot status query	`st?`

Telemetry Format

```
P:<position>,E:<error>,V:<velocity>,T:<target>,F:<fault>,M:<movi
```

Auto-Tune

Relay-based Ziegler-Nichols PID tuning. The motor oscillates around a target, measures ultimate gain (Ku) and period (Tu), then applies:

- Kp = 0.6 x Ku x 100 (capped at 10000)
- Ki = 1.2 x Ku / Tu (capped at 1000)
- Kd = 0.075 x Ku x Tu (capped at 10000)

Send `TUNE` to start. The motor performs relay oscillation (+/- 3000 steps/s relay velocity), and after 4+ full cycles computes PID gains and reports: `OK TUNE:Kp=...,Ki=...,Kd=...,amp=...,Tu=...`

Non-Volatile Config

All tuning parameters are auto-saved to the last flash page (512 KB, address 0x1D07F000) with 100 ms debounce. Integrity validated via magic word (0xBEADC0DE) and XOR checksum.

Hardware Testing

Test Setup

- **Motor:** NEMA17 stepper (200 steps/rev, 1.8 degrees/step)
- **Driver:** TB6600 with common-anode wiring
- **Microstep:** 1/32 (6400 steps/rev)
- **Encoder:** AS5047U magnetic encoder in ABI mode
- **Supply:** Regulated DC power supply at 24V and 31V
- **Belt drive:** G2 pulley, 20 teeth x 2 mm (40 mm/rev, 160 steps/mm)

- **Belt speed formula:** steps/s / 160

24V Test Data

Constant PID: Kp = 10, Ki = 5, Kd = 0. Varying MAXV, Accel, and coil current to characterize high-speed limits.

MAXV (steps/s)	Belt Speed (m/s)	RPM	Accel (steps/s^2)	Coil Current (A)	Result
250,000	1.56	2,344	3,200,000	2.5-2.7	Smooth with heating
250,000	1.56	2,344	3,200,000	3.0-3.2	Smooth with heating
250,000	1.56	2,344	3,200,000	2.8-2.9	Smooth with heating
250,000	1.56	2,344	3,200,000	2.5-2.7	Smooth with heating
230,000	1.44	2,156	3,300,000	2.0-2.2	Smooth with heating
200,000	1.25	1,875	1,900,000	1.5-1.7	Minute shuttering
200,000	1.25	1,875	190,000	1.0-1.2	**Stuck**

Conclusion: At 250k steps/s (3.2M steps/s^2 accel), minimum reliable coil current is 2.5-2.7A. At 230k/3.3M, 2-2.2A is sufficient. Below 2A with high speed, shuttering or stalling occurs.

31V Test Data

Systematic sweep of ACCEL, Vmax, and PID gains at 31V supply.

Voltage	Accel (steps/s^2)	Vmax (steps/s)	Kp	Ki	Kd	Measured Velocity	Target	Result
31V	3200000	300000	50	5	0	1670mm/s	25000	Smooth with heating
31V	3200000	300000	50	5	0	1780mm/s	25000	Smooth with heating
31V	3200000	300000	1000	5	0	1902mm/s	25000	Smooth with heating
31V	1400000	300000	1092	0	0	1445mm/s	25000	Smooth with heating
31V	1700000	300000	1092	46	0	1610mm/s	25000	Not Smooth
31V	1700000	300000	3000	46	0		25000	

Voltage	Accel (steps/s^2)	Vmax (steps/s)	Kp	Ki	Kd	Measured Velocity	Target	Result
						1543mm/s		Smooth With oscillation
31V	1800000	300000	3000	46	0	-	-	**Stuck**
31V	1700000	310000	3000	46	0	1470mm/s	-	**Stuck**
31V	2000000	240000	1050	46	0	1440mm/s	25000	Smooth with heating
31V	2700000	270000	1934	46	0	1669mm/s	25000	Smooth with heating
31V	2700000	280000	2121	46	0	1725mm/s	25000	Smooth with heating

Key Findings at 31V

- **300k Vmax / 3.2M Accel** works with moderate Kp (50-1000) and Ki = 5, Kd = 0 — best measured velocity 1902 mm/s
- **High Kp (> 2000) + Ki (> 0)** causes oscillation at 31V — Kp > 2000 is unstable with nonzero Ki
- **Vmax > 300k requires lower Accel** trade-off: 280k Vmax + 2.7M Accel works, 310k + 1.7M does not
- **Best reliable result:** 270k Vmax + 2.7M Accel, Kp = 2121, Ki = 46 — 1725 mm/s smooth
- **Kd = 0** in all tests — derivative gain not needed at 31V with proper Kp/Ki balance
- **Mid-band resonance** observed at position 3007 — higher Kd (500-2000) helps dampen; Ki > 10 causes integral windup

Voltage-Dependent Performance Limits

At 1/32 microstep (6400 steps/rev). Firmware accepts any positive value; the practical limit is set by the motor driver and supply voltage. Empirically determined guidelines (no load, idle condition):

Supply Voltage	MAXV (steps/s)	MAXV (RPM)	ACCEL (steps/s^2)	Notes
31V	280,000	2,625	2,700,000	Best reliable: 1725 mm/s, Kp = 2121, Ki = 46
24V	250,000	2,344	3,200,000	At >= 2.5A coil current, smooth

Supply Voltage	MAXV (steps/s)	MAXV (RPM)	ACCEL (steps/s^2)	Notes
				with heating
12V	50,000	469	5,000,000	Reduced max velocity

RPM formula: RPM = (steps/s x 60) / steps_per_rev where steps_per_rev = 6400 at 1/32.

All testing at 48 MHz SYSCLK — 10 MHz testing pending.

Schematics and PCB

Current Status

- **Schematic:** Complete in KiCad. All component symbols placed, wired with net labels, and PPS mapping documented.
- **PCB layout:** Pending — component placement, trace routing, thermal management, and DRC not yet started.

Power Architecture

```
USB-C VBUS (5V) -> LDO (MCP1825S / MCP1804) -> 3.3V rail (500
mA)
Motor PSU (9-30V) -> TMC2660 VM (direct)
                     -> Pre-regulator (optional for 5V CAN)

3.3V rail powers: PIC32MK, AS5047D, CAN transceiver
5V rail (optional): MCP2562 CAN transceiver (5V variant)
```

Rail	Source	Devices	Max Current
3.3V	LDO from VBUS	PIC32MK, AS5047D, CAN transceiver	~300 mA
5V (optional)	LDO from VBUS or motor PSU	MCP2562 (5V variant)	~70 mA
VM (9-30V)	External motor supply	TMC2660 motor driver	Depends on motor

PIC32MK0512MCJ064 Minimum Circuit

- **Decoupling:** 0.1 uF per VDD/VSS pair (3 pairs on 64-TQFP) within 3 mm of pin; one shared 10 uF bulk per chip
- **VCAP (pin 32):** 10 uF ceramic X7R, ESR < 5 ohms, routed directly with minimal loop
- **AVDD/AVSS:** 10 ohm isolate resistor + 10 uF bulk + 0.1 uF bypass

- **MCLR:** 10k ohm pull-up to 3.3V + 100 ohm series to ICSP header
- **Oscillator:** 24 MHz crystal + 18-22 pF load caps (optional — FRC+SPLL sufficient if no USB)

Pin Connections

TMC2660 (QFP-44) — SPI1 Bus

Pin	TMC2660	Connected To	Notes
4	CSN	RB4 (pin 21)	SPI chip select, active low
5	SCK	RB0 (pin 5)	SPI clock, PPS: RB0R = 8
6	SDI	RB1 (pin 6)	SPI data in (MOSI), PPS: RB1R = 7
7	SDO	RB2 (pin 7)	SPI data out (MISO), PPS: SDI1R = 5
25	ENABLE	RB13 (pin 42)	LOW = enabled
26	STEP	RB15 (pin 44)	OC1 PWM, PPS: RPB15R = 5
27	DIR	RB14 (pin 43)	LOW = CW, HIGH = CCW
8	REF	GND	Internal bandgap reference
31	TST	GND	Must be GND
9	VCC_IO	3.3V	Logic supply
35	VDD	3.3V	Analog supply
37	VCC	3.3V	Internal logic
39, 40	VM	Motor PSU (9-30V)	100-470 uF + 0.1 uF to GND
12	RSA	0.12-0.22 ohm to GND	Sense resistor A, Kelvin connection
22	SRB	0.12-0.22 ohm to GND	Sense resistor B, Kelvin connection
9	BRA	Same node as RSA	Bridge A return through sense resistor
25	BRB	Same node as SRB	

Pin	TMC2660	Connected To	Notes
			Bridge B return through sense resistor
32-34	CP/C1_B/C1_C	10 nF NPO	Charge pump capacitors
36	VS	VM	Supply voltage sense
35	VHS	VS via 100 nF	Bootstrap cap for high-side gate drive
13	5VOUT	470 nF to GND	Internal regulator, cap only
14, 15	OA1, OA2	Motor coil A	A+ and A-
18, 19	OB1, OB2	Motor coil B	B+ and B-
30	SG_TST	NC	Optional stallGuard2 output
37, 38	TST_ANA	NC	Do not connect

AS5047D (TSSOP-14) — Shared SPI1 Bus

Pin	AS5047D	Connected To	Notes
1	CSn	RB3 (pin 8)	Chip select, active low
2	CLK	RB0 (pin 5)	SPI clock (shared with TMC2660)
3	MISO	RB2 (pin 7)	SPI data out (shared)
4	MOSI	RB1 (pin 6)	SPI data in (shared)
5	VDD	3.3V + 0.1 uF	Main supply
6	VDD3V3	10 uF + 0.1 uF to GND	LDO output, do NOT drive externally
7	GND	GND	Ground
8	TEST	GND	Must be GND for normal operation
9	A	RPG6 (pin 53)	QEI channel A, PPS: QEA1R = 10
10	B	RPG7 (pin 54)	QEI channel B, PPS: QEB1R = 10
11	U	NC	UVW not used

Pin	AS5047D	Connected To	Notes
12	V	NC	UVW not used
13	W/PWM	NC	Not used
14	OffCompN	NC or 10k to 3.3V	Internal pull-up, safe floating

6-Pin ICSP Header

Pin	Signal	PIC32MK	Notes
1	MCLR	Pin 9 via 100 ohm	Reset
2	3.3V	VDD	Target power
3	GND	GND	Common ground
4	PGED1	RB10 (pin 24)	Programming data
5	PGEC1	RB11 (pin 25)	Programming clock
6	NC	—	Not connected

- Programmer: PKOB4 (serial BUR214320077)
- Command: `hwtool PKOB4 -p -i <hex file>`
- After MDB flash, chip may hold in reset until USB re-plugged; use `hwtool PKOB4 -p` for programming-only mode

Other Connections

Function	PIC32MK	Pin	Notes
U1TX	RPE0	16	Serial TX, PPS: RPE0R = 1
U1RX	RPG8	50	Serial RX, PPS: U1RXR = 10
USB D+	RGI0	51	Via 27 ohm + ESD diode
USB D-	RGI1	52	Via 27 ohm + ESD diode
CAN TX	RF2	26	PPS: RF2R = 11
CAN RX	RF3	27	PPS: C1RXR = 15
LED	RA10	4	Via 330 ohm to GND

TMC2660 Sense Resistor Kelvin Connection

```
        +-- Wide trace (1 mm) BRA ------+-- R_sense (0.12 ohm)
-- GND
TMC2660 |                                |
        +-- Thin trace (0.25 mm) SRA ---+
             (Kelvin — no motor current)
```

SRA and SRB carry ~0.5 mV signals. Use thin separate traces from the sense resistor pads back to the IC, not shared with the high-current BRA/BRB path.

Next Steps

1. **PCB layout** — route traces, place components, run DRC, generate Gerber files
2. **10 MHz clock migration** — after oscillator hardware is verified, recalculate:
   - U1BRG for 115200 baud at 10 MHz PB clock
   - Timer2 prescaler and PR2 for OC1 step generation
   - Timer3 PR3 for 1 kHz position control ISR
   - Core tick formulas (msleep, auto-tune Tu, telemetry timing)
3. **Re-test** at 10 MHz — repeat 24V and 31V characterization to verify no regression
4. **Fabrication** — send Gerber files to fab house, order components from BOM
5. **Assembly** — solder prototype boards, validate against test data
6. **CAN bus firmware** — implement CAN protocol for multi-axis communication

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Generated from firmware v6.2.0 test data. All testing performed at 48 MHz SYSCLK using SPLL from 8 MHz FRC (PLLIDIV = 2, PLLMULT = 24, PLLODIV = 2). 10 MHz target clock testing is pending.